

EENG250 ELECTRIC CIRCUITS I · CHAPTER 6

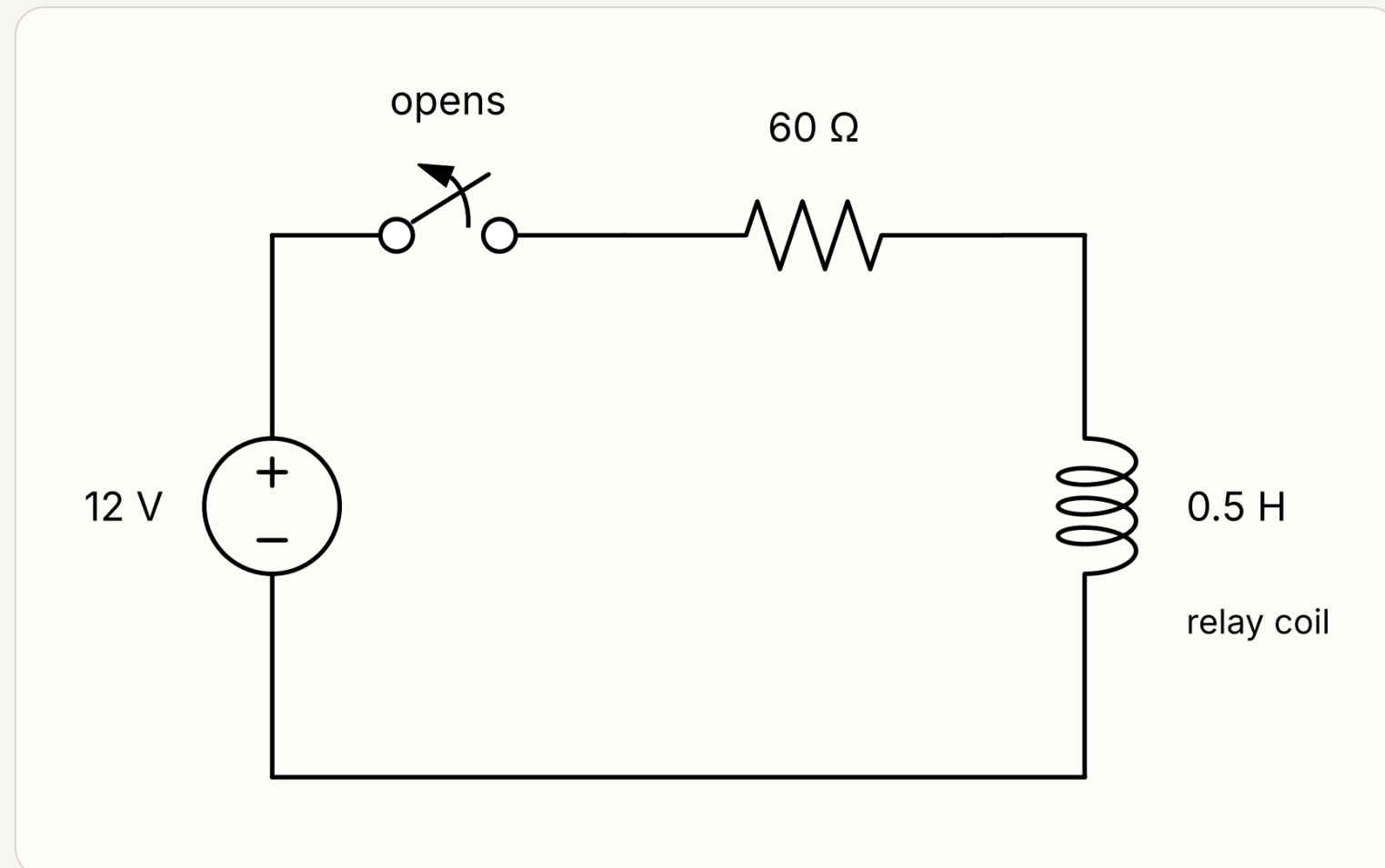
Inductors and Capacitors

Two elements that store energy and remember their past: an inductor's current and a capacitor's voltage never jump.

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PREDICT FIRST

A relay coil carries 0.2 A. The switch opens and the current falls to zero in about $10\ \mu\text{s}$. What voltage appears across the coil?



12 V

about 100 V

about 10 000 V

CHAPTER 6 GOALS

By the end of Chapter 6 you can

1 • The inductor

Find its voltage from its current, and its current from its voltage and initial current.

2 • The capacitor

Do the same for a capacitor's current and voltage.

3 • Power and energy

Find the stored energy, and say when an element stores or returns energy.

4 • Combinations

Combine elements with their initial values, and find the energy a load receives and the energy left trapped.

THE INDUCTOR

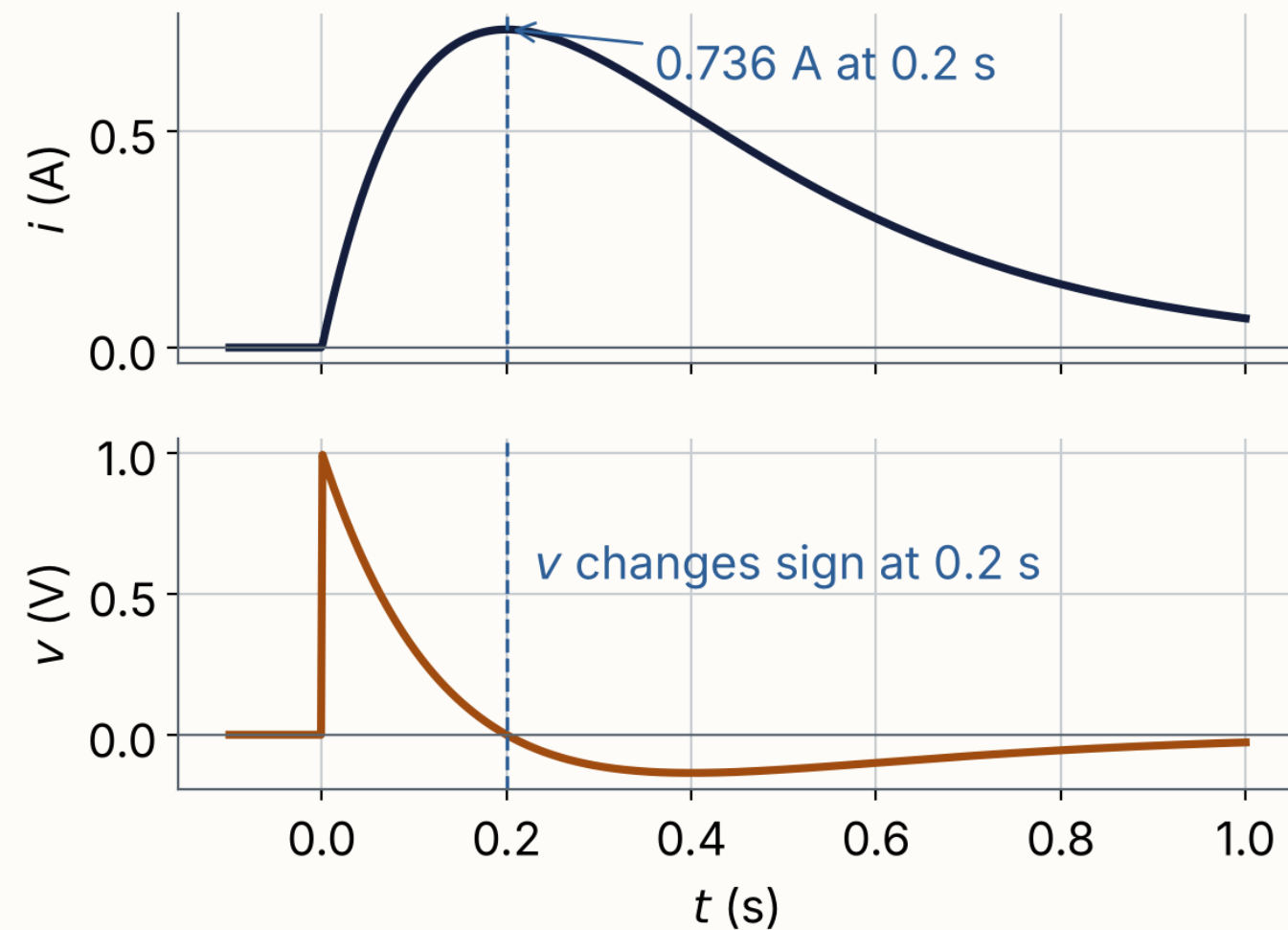
A changing current induces a voltage

$$v = L \frac{di}{dt}, \quad i(t) = \frac{1}{L} \int_{t_0}^t v d\tau + i(t_0)$$

- L in henrys (H). The current i is referenced in the direction of the voltage drop v.
- Constant current: the derivative is zero, so $v = 0$. In a dc circuit an inductor is a short circuit.
- The current cannot change instantly: a jump would need an infinite voltage. The voltage can jump.
- $i(t_0)$ carries the history; it is negative when the initial current flows against the arrow.

NILSSON EXAMPLE 6.1

Voltage from current: differentiate



A 100 mH inductor carries

$$i = 0, \quad t < 0; \quad i = 10te^{-5t} \text{ A}, \quad t > 0$$

$$\frac{di}{dt} = 10e^{-5t}(1 - 5t) = 0 \Rightarrow t = 0.2 \text{ s}, \quad i_{\max} = 2e^{-1} = 0.736 \text{ A}$$

$$v = L \frac{di}{dt} = (0.1)(10)e^{-5t}(1 - 5t) = e^{-5t}(1 - 5t) \text{ V}, \quad t > 0$$

v changes polarity at 0.2 s, when i peaks. v jumps from 0 to 1 V at t = 0; i does not jump.

NILSSON EXAMPLE 6.2

Current from voltage: integrate, then add $i(0)$

A voltage pulse is applied to a 100 mH inductor:

$$v = 20te^{-10t} \text{ V}, \quad t > 0; \quad i(0) = 0$$

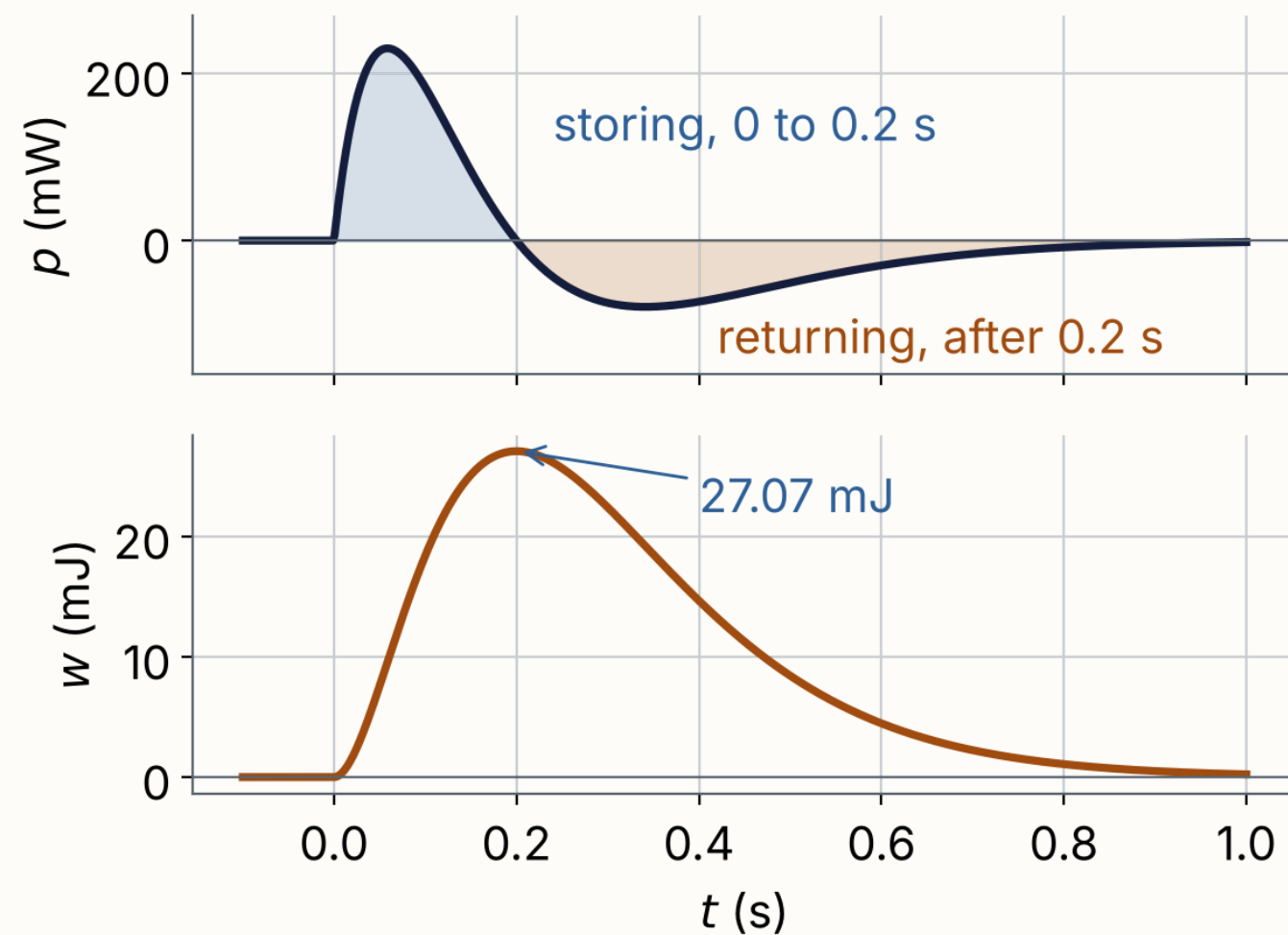
$$i = \frac{1}{0.1} \int_0^t 20xe^{-10x} dx + 0 = 2 (1 - 10te^{-10t} - e^{-10t}) \text{ A}$$

The integral needs integration by parts. Whatever method you use, check the result by differentiating it back:

$$\text{check: } 0.1 \frac{di}{dt} = 0.2 (-10e^{-10t} + 100te^{-10t} + 10e^{-10t}) = 20te^{-10t} \text{ V}$$

$$t \rightarrow \infty : \quad i \rightarrow 2 \text{ A}, \quad w = \frac{1}{2}(0.1)(2)^2 = 0.2 \text{ J} \quad \text{The current stays after the voltage has gone.}$$

Positive p stores energy; negative p returns it



$$p = vi = Li \frac{di}{dt}, \quad w = \frac{1}{2} Li^2$$

For Example 6.1:

$$p = vi = 10te^{-10t}(1 - 5t) \text{ W}, \quad w = \frac{1}{2} Li^2 = 5t^2e^{-10t} \text{ J}$$

$$w_{\max} = 5(0.2)^2e^{-2} = 27.07 \text{ mJ}$$

$$\int_0^{0.2} p \, dt = 27.07 \text{ mJ}, \quad \int_{0.2}^{\infty} p \, dt = -27.07 \text{ mJ}$$

All 27.07 mJ stored by 0.2 s is returned afterwards: an inductor dissipates nothing.

THE CAPACITOR

A changing voltage drives a current

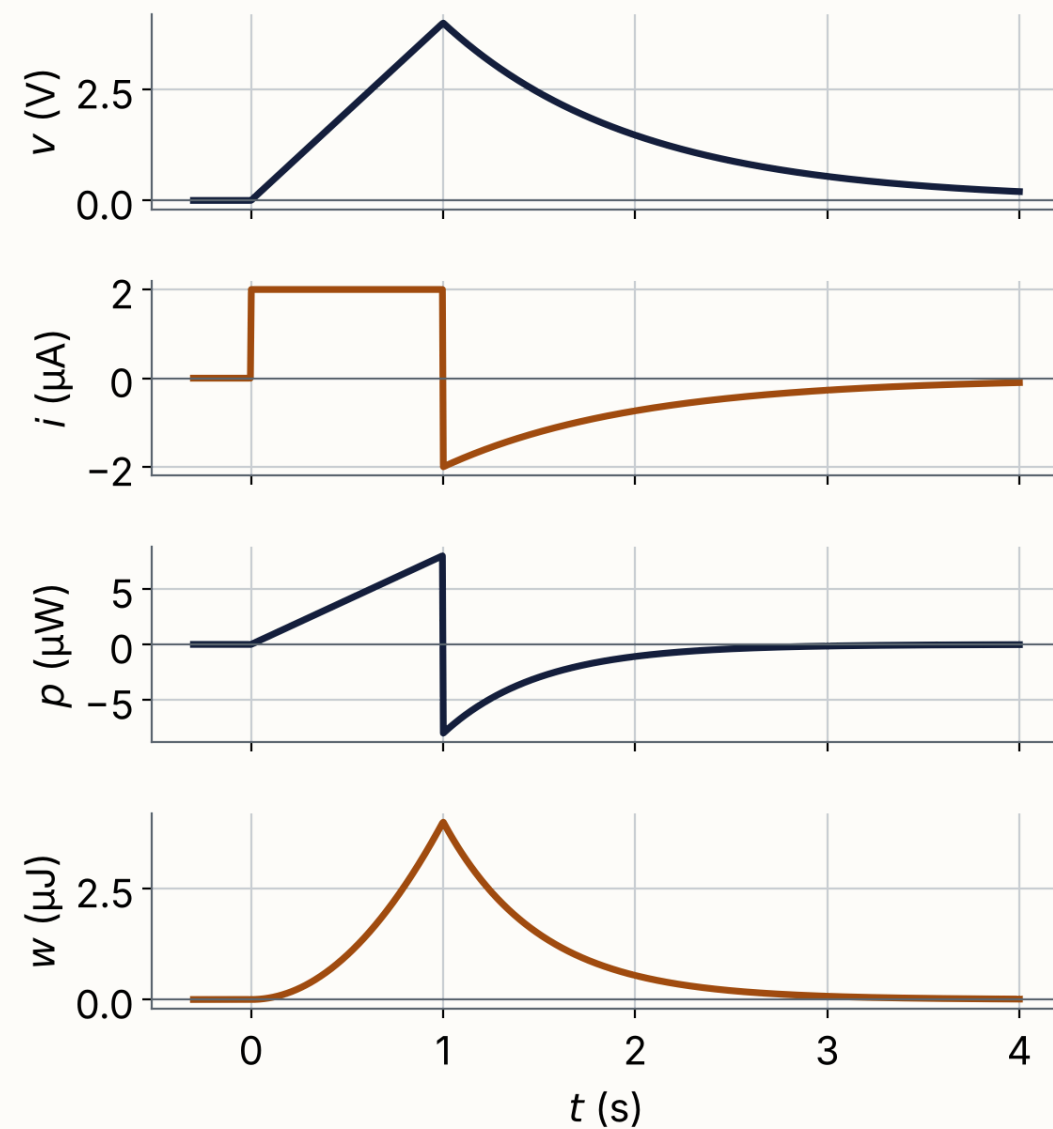
$$i = C \frac{dv}{dt}, \quad v(t) = \frac{1}{C} \int_{t_0}^t i d\tau + v(t_0)$$

$$p = vi = Cv \frac{dv}{dt}, \quad w = \frac{1}{2} C v^2$$

- C in farads (F); circuits use μF , nF and pF.
- Constant voltage: the derivative is zero, so $i = 0$. In a dc circuit a capacitor is an open circuit.
- The voltage cannot change instantly. The current can jump.

NILSSON EXAMPLE 6.4

A voltage pulse across a $0.5 \mu\text{F}$ capacitor



$$v = 4t \text{ V}, \quad 0 \leq t \leq 1 \text{ s}; \quad v = 4e^{-(t-1)} \text{ V}, \quad t \geq 1 \text{ s}$$

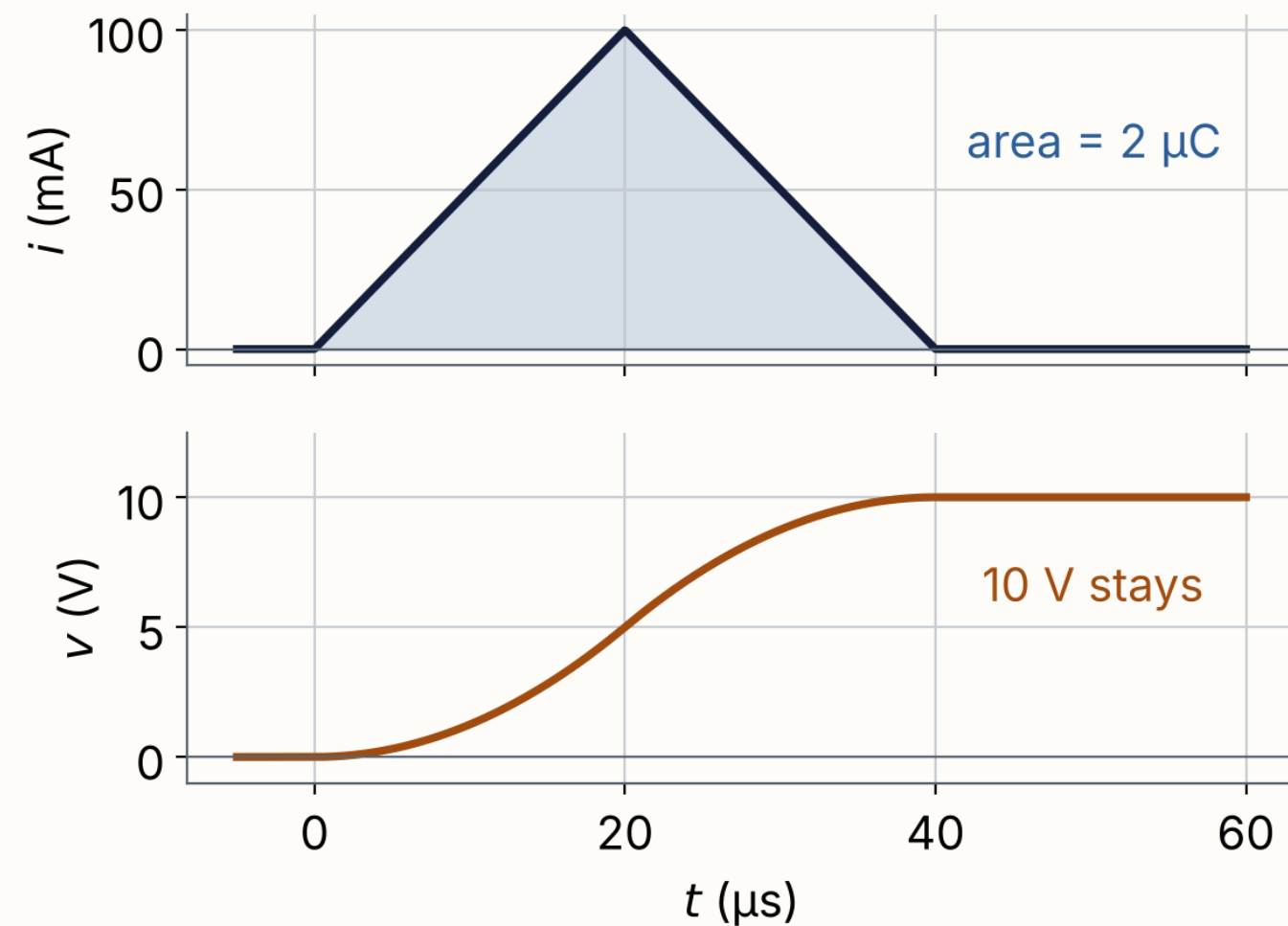
$$i = C \frac{dv}{dt} = \begin{cases} (0.5 \mu)(4) = 2 \mu\text{A}, & 0 < t < 1 \text{ s} \\ (0.5 \mu)(-4e^{-(t-1)}) = -2e^{-(t-1)} \mu\text{A}, & t > 1 \text{ s} \end{cases}$$

$$\int_0^1 p \, dt = \int_0^1 8t \, dt = 4 \mu\text{J}, \quad \int_1^\infty p \, dt = -4 \mu\text{J}$$

Energy is stored from 0 to 1 s and returned after 1 s. The current jumps at $t = 1$ s; the voltage does not.

NILSSON EXAMPLE 6.5

A current pulse leaves a voltage behind



An uncharged 0.2 μF capacitor receives a triangular current pulse.

The voltage is the running area under i , divided by C :

$$q = \text{area} = \frac{1}{2}(40 \mu\text{s})(0.1 \text{ A}) = 2 \mu\text{C}, \quad v = \frac{q}{C} = \frac{2 \mu\text{C}}{0.2 \mu\text{F}} = 10 \text{ V}$$

After the pulse no current flows, so the voltage stops changing: the charge, and the 10 V, stay.

PREDICT FIRST

A 2 μF capacitor in a dc circuit, long after switching, has 10 V across it. What current flows in it?

20 μA

10 μA

0

0. The voltage is constant, so no current flows: in dc a capacitor is an open circuit. It still stores energy.

$$i = C \frac{dv}{dt} = 0, \quad w = \frac{1}{2}(2 \mu\text{F})(10)^2 = 100 \mu\text{J}$$

20 μA comes from multiplying C by v, and 10 μA from dividing the energy by v; both forget that the current depends on how fast the voltage changes.

INDUCTOR AND CAPACITOR SIDE BY SIDE

Swap v and i, L and C

	Inductor	Capacitor
In a dc circuit	Short circuit ($v = 0$)	Open circuit ($i = 0$)
Cannot change instantly	Its current	Its voltage
In series	Inductances add	Reciprocals add
In parallel	Reciprocals add	Capacitances add
Initial values add	Currents of parallel inductors (KCL)	Voltages of series capacitors (KVL)

SERIES AND PARALLEL COMBINATIONS

Inductors combine like resistors; capacitors the other way round

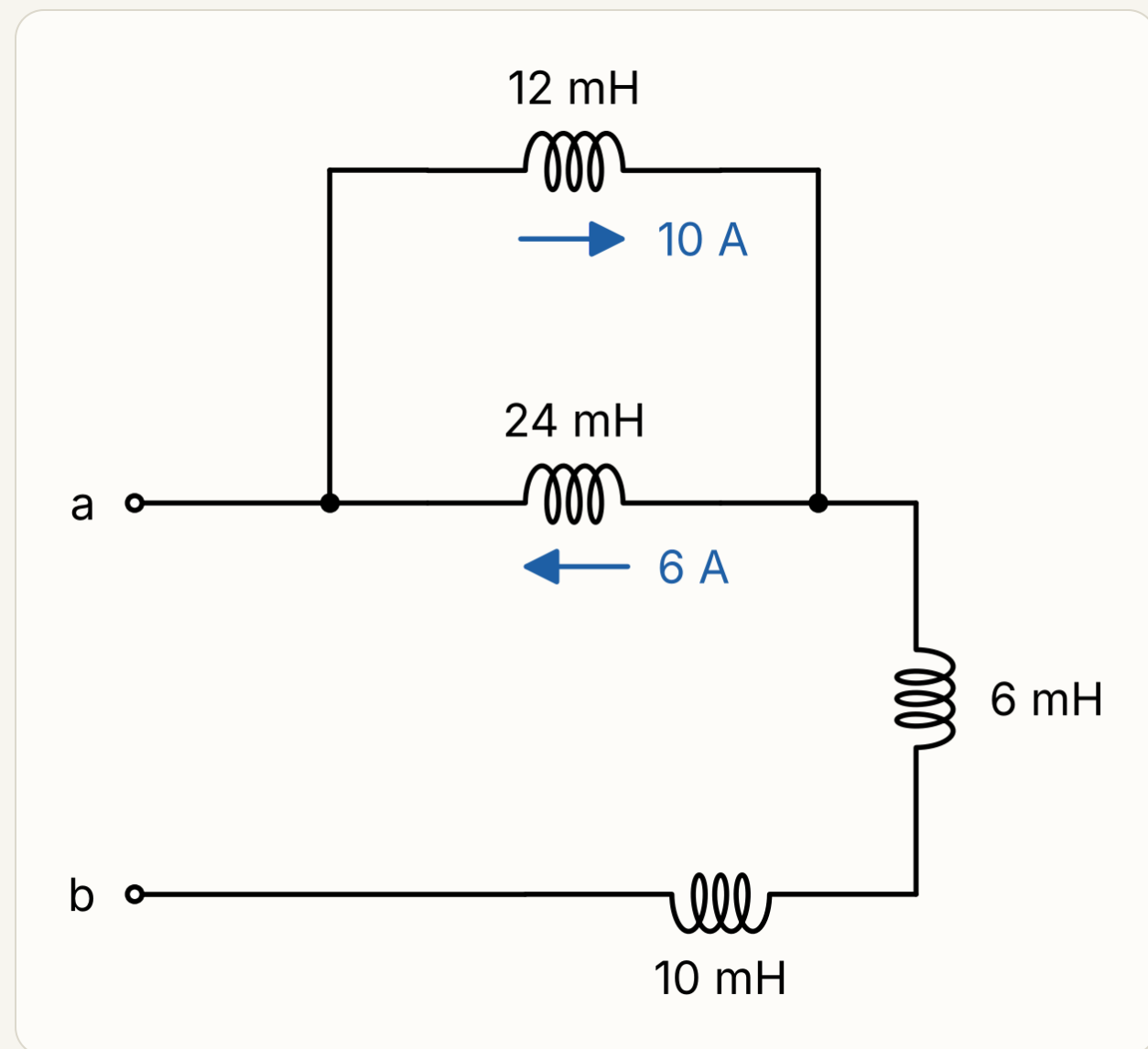
$$\text{series: } L_{eq} = L_1 + L_2 + \cdots \quad \text{parallel: } \frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2} + \cdots, \quad i(0) = i_1(0) + i_2(0) + \cdots$$

$$\text{series: } \frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \cdots, \quad v(0) = v_1(0) + v_2(0) + \cdots \quad \text{parallel: } C_{eq} = C_1 + C_2 + \cdots$$

Parallel inductors: the initial currents add (KCL). Series capacitors: the initial voltages add along the path (KVL). Watch the directions.

NILSSON EXAMPLE 6.6

Equivalent inductance and its initial current

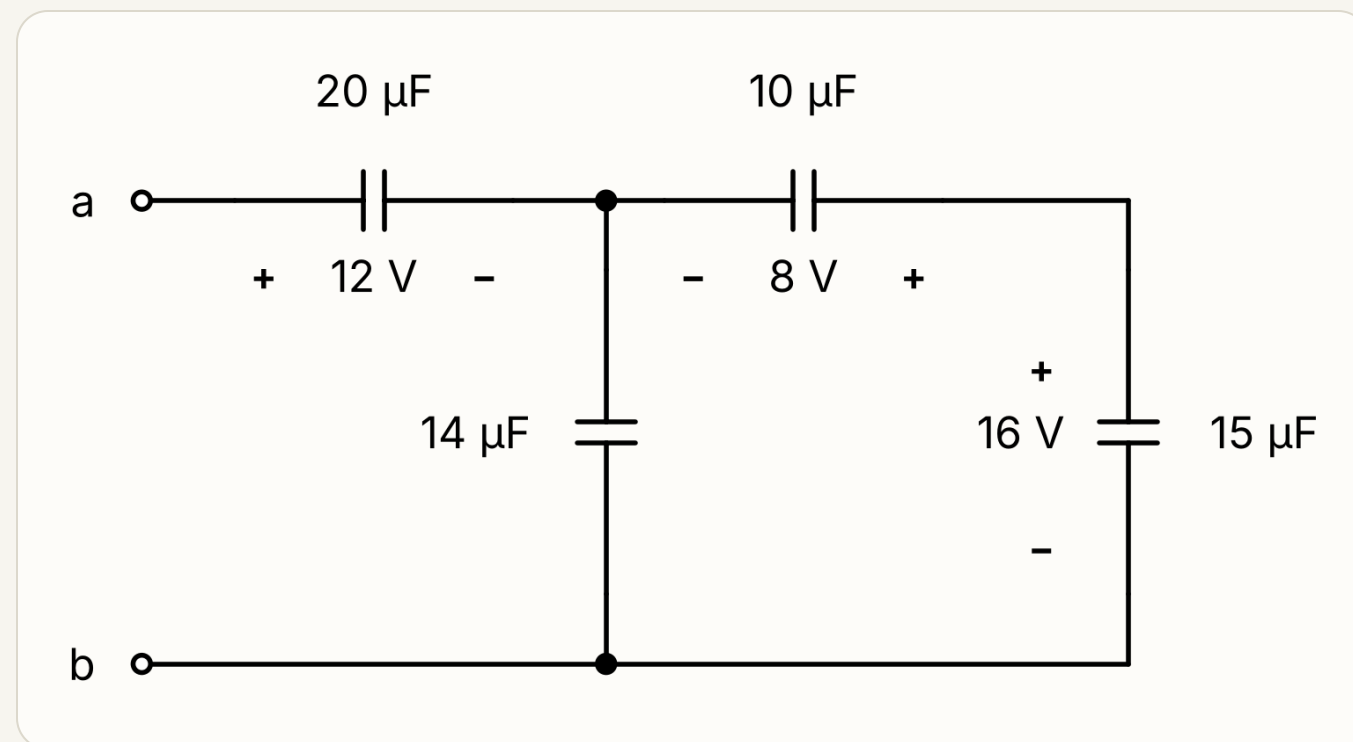


$$L_{eq} = \left(\frac{1}{12} + \frac{1}{24} \right)^{-1} + 6 + 10 = 8 + 6 + 10 = 24 \text{ mH}, \quad i(0) = 10 - 6 = 4 \text{ A}$$

The current entering at a is the current entering the node at the left of the parallel pair. KCL with directions: 10 A leaves through the 12 mH, 6 A arrives through the 24 mH.

NILSSON EXAMPLE 6.7

Equivalent capacitance and its initial voltage



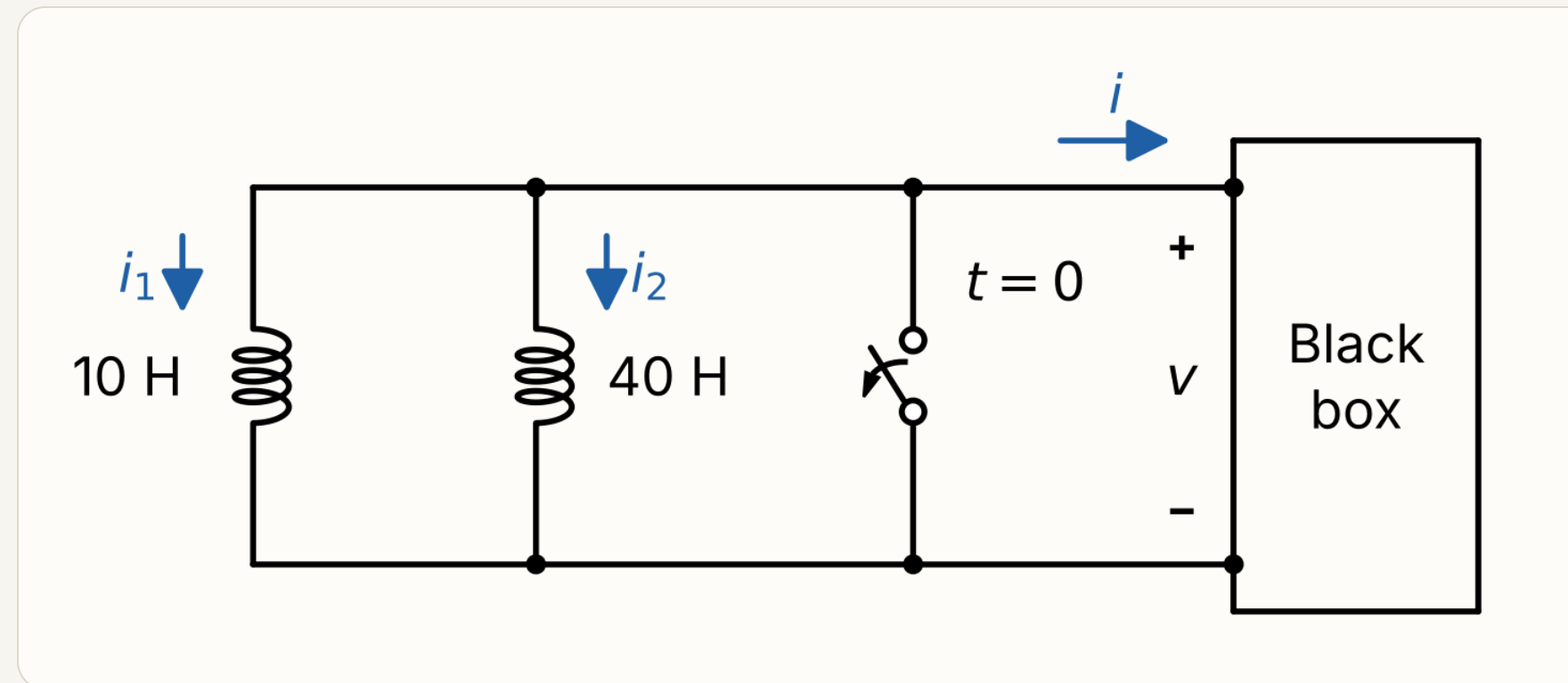
10 μF in series with 15 μF : $\left(\frac{1}{10} + \frac{1}{15}\right)^{-1} = 6 \mu\text{F}$

$$C_{eq} = \left(\frac{1}{20} + \frac{1}{14 + 6}\right)^{-1} = 10 \mu\text{F}, \quad v(0) = 12 - 8 + 16 = 20 \text{ V}$$

KVL along the outside path from a to b, with each capacitor's polarity.

A BLACK BOX DRAWS ON STORED ENERGY

Find the currents first



$$i_1(0) = -3 \text{ A}, \quad i_2(0) = 2 \text{ A}; \quad v = 80e^{-10t} \text{ V}, \quad t > 0$$

Until $t = 0$ the closed switch carries the total current. When it opens, that current flows into the box: $i = -(i_1 + i_2)$.

$$L_{eq} = \frac{(10)(40)}{10 + 40} = 8 \text{ H}, \quad i(0) = -[i_1(0) + i_2(0)] = -(-3 + 2) = 1 \text{ A}$$

$$i_1 = -3 + \frac{1}{10} \int_0^t 80e^{-10x} dx = -2.2 - 0.8e^{-10t} \text{ A}, \quad i_2 = 2 + \frac{1}{40} \int_0^t 80e^{-10x} dx = 2.2 - 0.2e^{-10t} \text{ A}$$

DELIVERED AND TRAPPED ENERGY

Not all of the stored energy reaches the box

$$w(0) = \frac{1}{2}(10)(3)^2 + \frac{1}{2}(40)(2)^2 = 125 \text{ J}, \quad w(\infty) = \frac{1}{2}(10)(2.2)^2 + \frac{1}{2}(40)(2.2)^2 = 121 \text{ J}$$

$$w_{\text{delivered}} = 125 - 121 = 4 \text{ J} = \frac{1}{2}(8)(1)^2$$

After a long time $i_1 = -2.2 \text{ A}$ and $i_2 = 2.2 \text{ A}$: a current circulates in the inductor loop and keeps 121 J trapped.

Shortcut

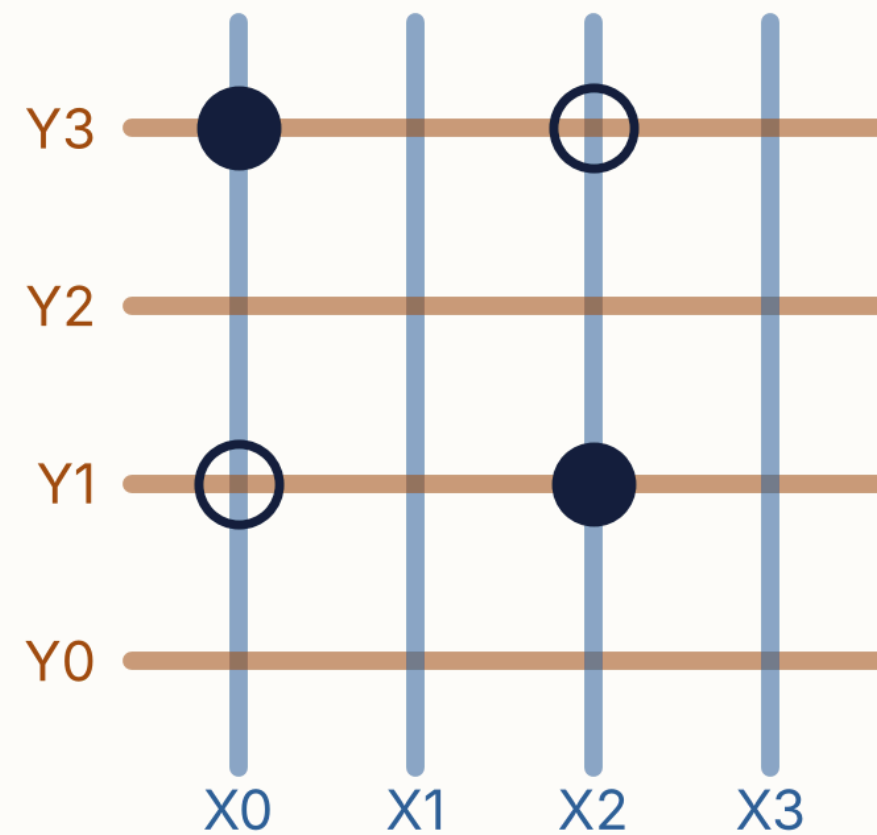
$$w_{\text{box}} = \frac{1}{2}L_{eq} i(0)^2$$

When the box draws the total current to zero, it receives the energy of the equivalent inductor; the rest stays trapped.

PRACTICAL PERSPECTIVE: CAPACITIVE TOUCH SCREENS

A touch changes the measured capacitance

● touch ○ ghost (self-capacitance)



Self-capacitance: each electrode has a parasitic capacitance C_p to ground, and a touch adds C_t in parallel: $C_{tx} = C_t + C_p$.

Self-capacitance: the controller learns which X and which Y electrodes changed. Two touches light four crossings, two of them ghosts.

Mutual capacitance: the controller measures the capacitance at each X–Y crossing; a touch changes it there, so every touch is located exactly.

CHECKING AN AI ANSWER

An AI found the initial current in Example 6.6's equivalent inductor like this. What is wrong?

$$i(0) = 10 + 6 = 16 \text{ A}$$

Correct:

$$\text{KCL at the node left of the 24 mH, currents leaving: } -i + 10 - 6 = 0 \Rightarrow i(0) = 4 \text{ A}$$

It added the two magnitudes and ignored the arrows: the 6 A flows the other way. KCL with directions gives 4 A, as Nilsson finds.

In a 2025 study of Gemini 2.5 Pro on undergraduate circuit problems, misread source polarities caused about a third of its wrong answers, and misread current directions nearly as many.

CHECK EVERY ANSWER

Three checks, no second solution needed

$$i_L(0^+) = i_L(0^-), \quad v_C(0^+) = v_C(0^-), \quad w(0) = w_{\text{delivered}} + w_{\text{trapped}}$$

Put $t = 0$ in

Your expression must return the given initial current or voltage, with its sign.

Differentiate back

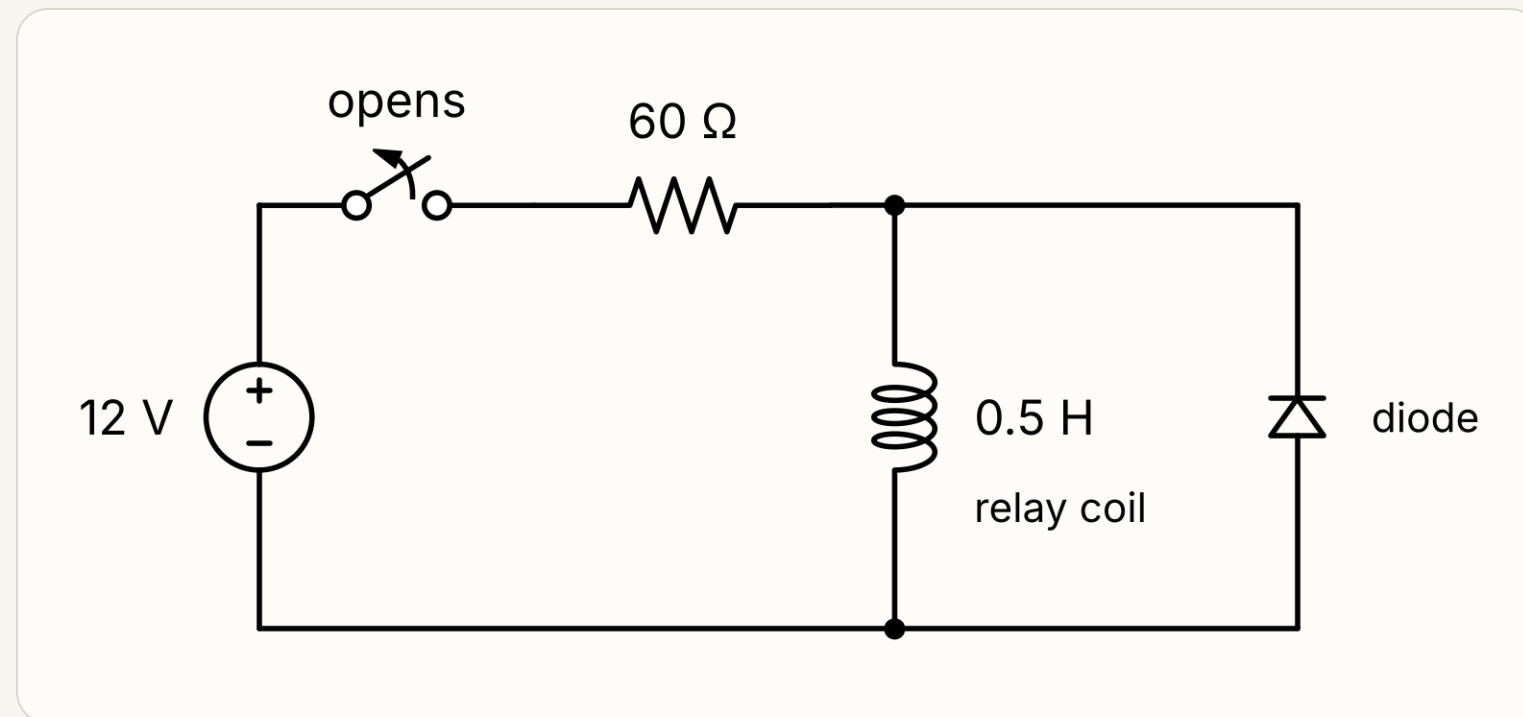
The element law must return the voltage or current you were given.

Balance the energy

Initial energy = energy delivered + energy trapped.

BACK TO THE RELAY

The coil's current needs somewhere to go



$$v = L \frac{di}{dt} \approx (0.5) \frac{0.2 \text{ A}}{10 \mu\text{s}} = 10\,000 \text{ V}$$

$$w = \frac{1}{2} Li^2 = \frac{1}{2} (0.5)(0.2)^2 = 10 \text{ mJ}$$

A diode across the coil lets the current keep flowing round the coil–diode loop after the switch opens. The 10 mJ dies away there instead of arcing across the contacts.

WHAT COMES NEXT

Practise openly, then show it on your own

In class

Booklet Problems 6.1 to 6.10.
Problem 6.11 is homework.

At home

The 16-item practice sheet. AI
tools allowed; not marked.

Week 12

15-minute quiz, no devices.
Every question is a twin of a
practice-sheet question.

Chapter 7: an inductor or a capacitor with a resistor. The continuity rules give the starting values, and the dc rules give the final ones.

CHECKLIST

You can do these without help

- Find an inductor's voltage from its current, and its current from its voltage and initial current
- Do the same for a capacitor's current and voltage
- Find the power and stored energy, and say when the element stores or returns energy
- State the dc behaviour of each element and which quantity cannot change instantly
- Combine inductors or capacitors in series and parallel, with their initial currents or voltages
- Find the energy delivered to a black box and the energy trapped in the elements